



Thinking outside the box

of embodied cognition – a field of science that believes that human intelligence can't be boiled down to the functioning of the brain, and that higher intelligence is impossible without perception, movement and visualisation.

This approach quickly led to the emergence of computational intelligence, which is a massively popular branch of AI today, which is essentially an amalgamation of the best aspects of pre-existing techniques. It borrows from “neat” theories, “scruffy” theories, fuzzy logic systems and even neural networks.

The most important breakthrough in computational intelligence came in 1986, when American psychologists David Rumelhart and James McClelland wrote a paper called *Parallel Distributed Processing: Explorations in the Microstructure of Cognition*. In it, the duo described in great detail the inner workings of the human brain in association with perception and created computer simulations of perception. This was the first time that scientists could test neural processing, thus making AI more true-to-life.

Essentially, computational intelligence took the field of AI outside of “the human brain” and made it about “the human mind” in all its